Who Loses Independence Next? New Daily-Living Limitations in Older Europeans

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In this paper we study which adults aged 65 and older who still manage all daily activities on their own develop a first limitation by the next SHARE survey wave. Knowing who is at risk can help target prevention before independence is lost.

Background

Europe's population is ageing, and with it grows the number of people who need help in everyday life. A common way to measure this is the Activities of Daily Living (ADL) score: whether a person has difficulty dressing, bathing, eating, getting in or out of bed, or using the toilet [1]. A first ADL limitation is often the moment when an older person starts to depend on care from family or professionals. This risk climbs steeply with age: among SHARE respondents who were still independent, 3.5% of those aged 50–59 reported a new ADL limitation by the next survey wave, compared with 9.0% at 70–79, 18.9% at 80–89 and 35.9% at 90+ (Figure 1). Which older adults who are still fully independent today will report such a limitation two or three years later? Is it mainly a question of age, or do health, mood, strength and early functional difficulties tell us more? And can a simple, transparent model rank people by their risk well enough to be useful? This paper follows independent adults aged 65+ across the waves of the Survey of Health, Ageing and Retirement in Europe (SHARE) to answer these questions.

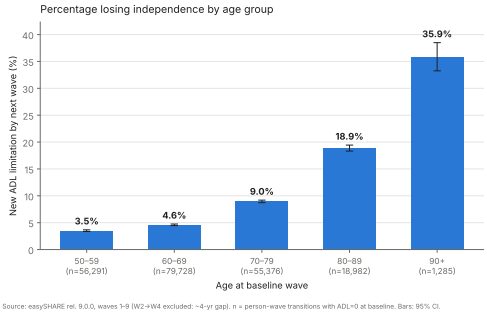


Figure 1: Share of independent respondents (ADL = 0) who lose independence by the next SHARE wave, by age group (all ages 50+; bars: 95% CI).

Data & Methods

We use easySHARE release 9.0.0 [2, 3, 5], a harmonised version of SHARE with 488,400 interviews and 108 variables from respondents in many European countries and Israel. For every respondent aged 65+ without any ADL limitation, we looked at the next survey wave in which they were interviewed again (the life-history wave 3 has no ADL data and was skipped). This gave 115,155 "before–after" pairs, of which 9.6% reported a new ADL limitation at follow-up, usually two years later.

Cleaning the data turned out to be an important step. easySHARE stores missing answers as negative numbers such as −10 or −15. Left untouched, they look like real values: the depression score, which ranges from 0 to 12, had a mean of 0.28 instead of 2.29 (Figure 2). After removing these codes, depression, BMI and other variables showed associations with future decline that had been hidden before.

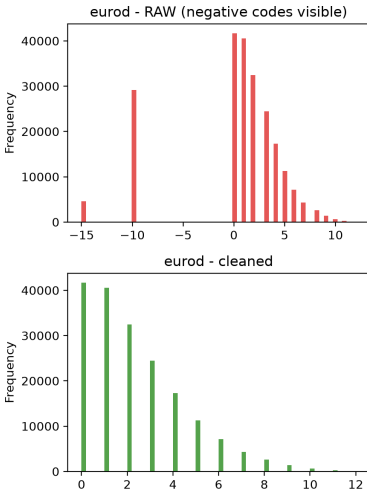


Figure 2: Depression score before (top) and after (bottom) removing missing-value codes.

As predictors we used what was known about each person at the first interview, in three groups: basic health (age, sex, self-rated health, chronic diseases, grip strength, depression, BMI, income and memory), social and lifestyle context (quality of life, physical activity, help received or given, living with a partner, education, doctor visits and hospital stays), and early functional difficulties (problems with walking and climbing stairs, with tasks such as getting up from a chair, and with instrumental tasks such as shopping or managing money). We first described decline rates across groups, then used principal component analysis (PCA) to see how strongly the predictors overlap. Finally we built a logistic regression model trained on earlier waves and

tested on the two most recent wave pairs, and compared it with an elastic-net logistic regression and a random forest.

Results

Age shows the clearest pattern: 5.1% of people aged 65–69 developed a new ADL limitation by the next wave, compared with 15.3% at 80–84 and 25.6% at 85+ (Figure 3). Self-rated health shows a similar gradient: respondents who rated their health as poor had the highest risk, far above those who rated it as excellent. Physical activity matters as well: people who hardly ever did vigorous activity declined in 14.4% of cases, compared with 5.0% among those active more than once a week. Women had a slightly higher raw rate than men (10.2% vs. 8.9%), but once age and health were taken into account, their risk was lower.

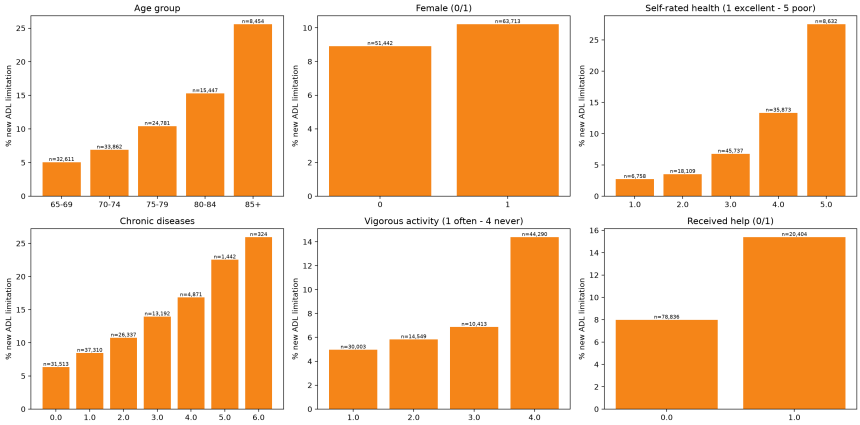


Figure 3: Share of respondents with a new ADL limitation at the next wave, by age group, sex, self-rated health, number of chronic diseases, vigorous activity and help received.

Do these signals all measure the same thing, namely general frailty? PCA suggests they do not. The first component does combine poor self-rated health, depression, weak grip, poor memory and higher age into a kind of "frailty" dimension, but only two components have an eigenvalue above 1, and four components still explain only two thirds of the variance (Figure 4). The predictors are only weakly correlated, so compressing them into a few components lost information and did not improve prediction. We therefore kept the original variables, which also lets us interpret each of them directly.

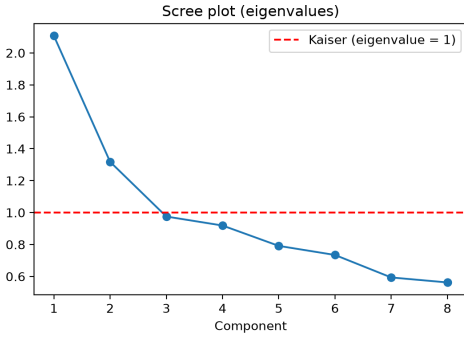


Figure 4: PCA of the eight main health predictors: eigenvalue per component.

The most informative signals were early functional difficulties. Among people with no mobility problems, 6.3% developed an ADL limitation; among those with three or more, 34.8% did. Adding these indicators improved the model's ability to separate future cases from non-cases clearly (AUC from 0.750 to 0.784), while switching to an elastic net or a random forest with the same information changed almost nothing. In other words, *what* we know about a person matters more than *how* we model it. In the final model, each additional difficulty with instrumental tasks raises the odds of a new limitation by 44%, each step worse in self-rated health by 34% and five more years of age by 29%; difficulties with large-muscle tasks also raise the risk. A stronger grip (odds 10% lower per 5 kg), better memory and higher quality of life lower it.

The model ranks people well (Figure 5): in the tenth of respondents with the lowest predicted risk, only 1.0% developed a limitation, compared with 32.3% in the highest tenth. The 20% of people the model considers most at risk account for 55.6% of all new cases. Its predicted risks also match the observed rates closely. However, it is not precise enough for individual diagnosis: when it flags 13% of respondents as high risk, about three in ten of them actually decline. The results remained stable when we tested different time periods, including the COVID-19 years, and when each person was counted only once.

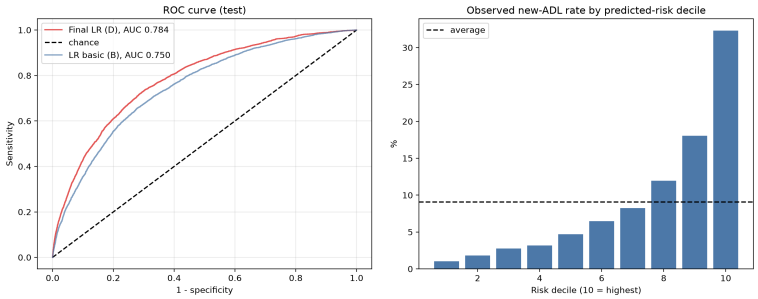


Figure 5: Final model on the most recent waves. Left: ROC curve with (red) and without (blue) functional indicators. Right: observed share of new ADL limitations in each tenth of predicted risk.

Conclusion

Our analysis of 115,155 SHARE follow-ups shows that a first loss of independence in old age is foreseeable to a useful degree. Age and self-rated health are strong warning signs, but early functional difficulties, such as trouble walking 100 metres, getting up from a chair or managing money, are the most informative ones. A transparent model built on these signals can separate low-risk from high-risk older adults and could help focus prevention, for example strength and mobility programmes, on the people who need it most. Two lessons came from the process itself: careful data cleaning changed our conclusions about depression and BMI, and more complex methods (PCA, an elastic net, a random forest) did not beat a simple model with good information. Because ADL is self-reported and functional difficulties are close to the outcome, the model should be seen as a tool for risk stratification, not diagnosis. Further work could look for earlier, non-functional warning signs and test the model on data outside SHARE. The code for this analysis is available online [4].

Acknowledgements

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